ICC 113 | Engineering in Society | Stellenbosch University

Sustainable Development Goal 1: No Poverty

POVERTY IN SOUTH AFRICA: UNEMPLOYMENT AS A SYSTEMIC DRIVER

OF INEQUALITY AND ECONOMIC EXCLUSION

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TABLE OF CONTENT

1. INTRODUCTION

Group

Engineering is increasingly recognised as a discipline that must engage not only with technical problems, but also with the complex social, economic and environmental challenges that define modern society. The United Nations Sustainable Development Goals (SDGs) provide a globally agreed framework for addressing these challenges, with SDG 1 specifically targeting the elimination of poverty in all its forms everywhere (United Nations, 2024). In South Africa, poverty remains a critical and deeply entrenched issue, directly linked to persistently high levels of unemployment that limit household income, restrict access to basic services, undermine skills development and reduce long-term economic participation.

South Africa's unemployment crisis is significant in both scale and persistence. The official unemployment rate stood at 32.9% in the first quarter of 2024, with 8.2 million people without work; the expanded unemployment rate, which includes discouraged work-seekers, reached 41.9%; and youth unemployment for those aged 15–24 was recorded at 59.7% (Statistics South Africa, 2024a). These figures confirm that unemployment is not a temporary fluctuation but a structural problem with consequences that extend far beyond the labour market, affecting health, food security, education, community stability and future economic growth.

This report investigates poverty in South Africa through a focused analysis of unemployment as its primary systemic driver. Using the Engineering Graduate Attributes (GA 1–7 and GA 10) as investigative lenses, the report examines the complexity of the problem, the knowledge base required to understand it, the tools and methods relevant to its investigation, and the ethical responsibilities that engineers carry when engaging with such challenges. The purpose of the report is not to propose a single solution, but to conduct a rigorous, evidence-based investigation into why unemployment matters, who it affects, and what factors must be considered from a holistic engineering perspective.

2. GA 1: COMPLEX PROBLEM SOLVING

Henry Venter – 30011906

Graduate Attribute 1 requires engineers to identify and engage with complex, open-ended problems. Unemployment in South Africa qualifies as such a problem because it is shaped by interacting economic, social, infrastructural, historical and educational factors that cannot be resolved through a single technical intervention. Understanding unemployment requires a systems-level analysis that reveals how multiple drivers reinforce one another and why the problem persists despite existing interventions.

As illustrated in Figure 1, poverty levels vary significantly across South Africa's provinces when measured by income per capita relative to Gauteng. Eastern Cape and Limpopo provinces show income levels approximately half that of Gauteng, confirming that poverty and unemployment are geographically concentrated in areas where infrastructure, transport, education and economic opportunity are most limited. Rural provinces and informal urban settlements face compounding disadvantages that exclude residents from labour markets and formal economic participation.

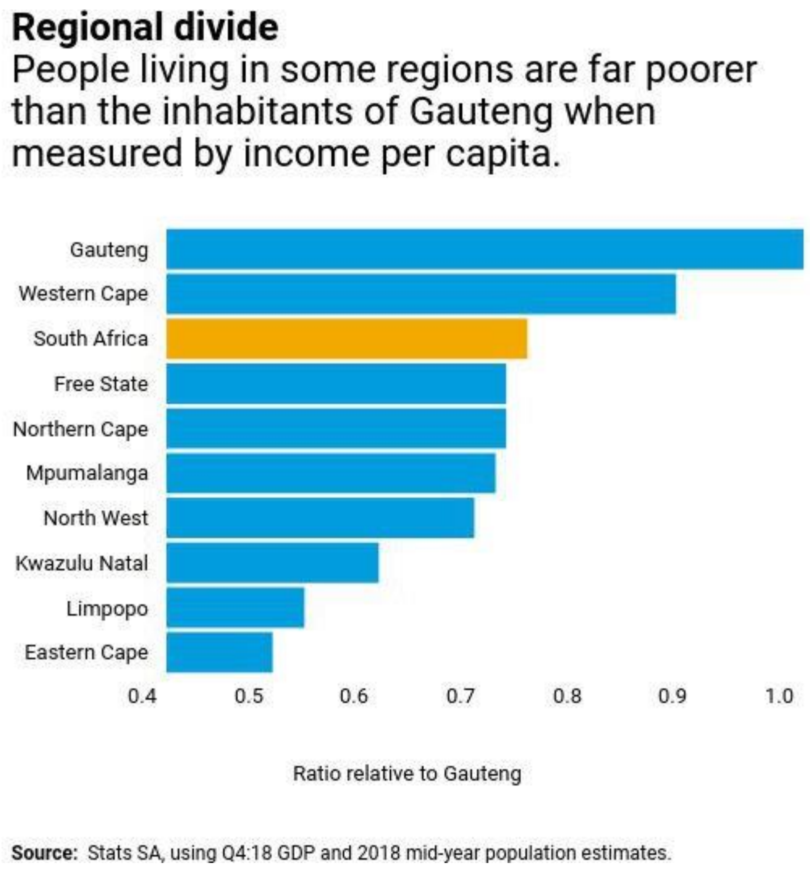


Figure 1: Regional poverty ratios in South Africa relative to Gauteng, measured by income per capita. (Source: Statistics South Africa, 2024b)

At the national level, Statistics South Africa (2024a) reported that 8.2 million people were unemployed in Q1 2024, with an official rate of 32.9%. Figure 2 shows how South Africa’s overall and youth unemployment rates dramatically exceed emerging-market averages, with youth unemployment exceeding 50% of the labour force at the time of data collection (International Monetary Fund, 2020). This international comparison confirms that structural conditions within South Africa are unusually severe and that the problem requires sustained, systems-level analysis rather than short-term responses.

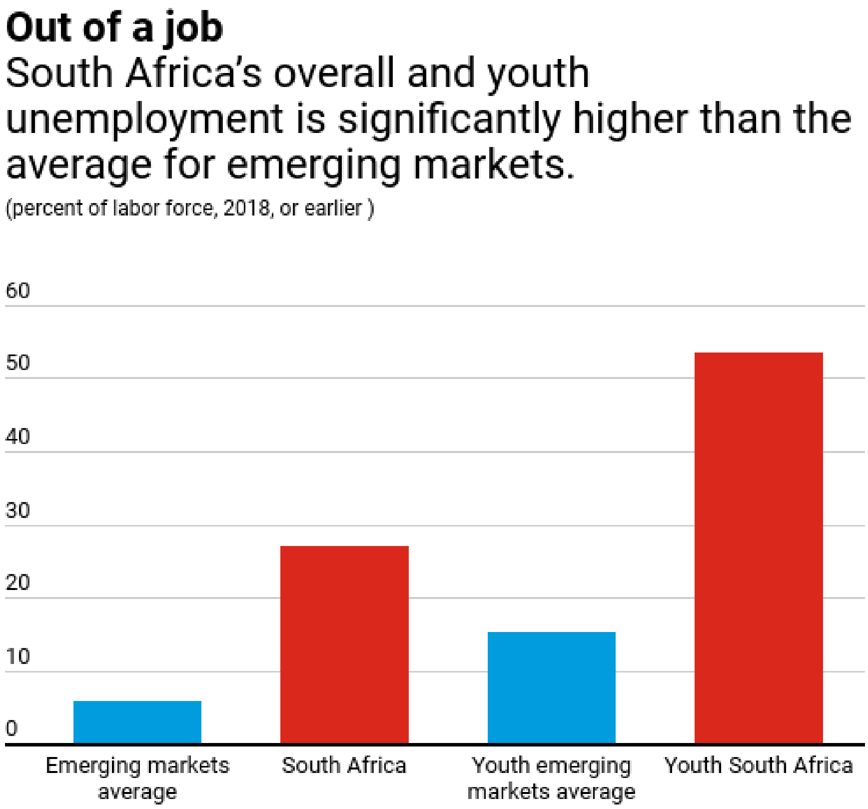


Figure 2: South Africa’s overall and youth unemployment compared with emerging-market averages. (Source: International Monetary Fund, 2020)

Complex problem solving in engineering means identifying interconnected causes rather than treating symptoms in isolation. Poor transport networks disconnect workers from job centres. Unreliable energy limits business investment. Inadequate education perpetuates skills shortages. Weak municipal infrastructure discourages economic activity in disadvantaged areas. GA 1 demonstrates that engineers contribute by mapping these interacting constraints, comparing alternatives and providing the evidence base for decisions that reduce exclusion and strengthen access to economic participation.

Unemployment in South Africa is not a single labour-market failure. It is a system of reinforcing exclusions across infrastructure, education, transport and economic access.

3. GA 2: ENGINEERING KNOWLEDGE

Beldon Kahn – 30056756

Engineering knowledge is not limited to technical or physical design. It encompasses mathematical reasoning, systems thinking, data analysis and the integration of socio-economic, historical and environmental understanding into evidence-based investigation. In the context of poverty and unemployment, engineering knowledge provides the analytical tools needed to understand why unemployment persists and how physical and institutional infrastructure shapes access to opportunity.

Mathematically, engineering knowledge is applied through statistical analysis of unemployment trends, poverty rates, income distributions and labour-market participation data. These tools allow researchers to identify patterns over time, compare provinces and test relationships between variables. The link between education and employment is clearly evidenced in the data: in Q1 2024, individuals without a matric qualification faced an unemployment rate of 39.1%, while graduates faced a rate of 11.8% (Statistics South Africa, 2024a). This quantitative relationship confirms that skills gaps produced by historically unequal education systems feed directly into unemployment.

South Africa's historical context is essential to understanding the current crisis. The Apartheid system deliberately created unequal access to education, housing, transport and employment, and these structural inequalities continue to shape labour-market outcomes. The International Monetary Fund (2020) confirms that South Africa's inequality is reinforced by skewed income distribution, unequal access to opportunity and regional disparities, with unemployment as the primary mechanism through which historical disadvantage persists into the present.

Engineering knowledge connects these social realities to physical systems. Poor road design increases the cost of job-seeking. Unreliable energy reduces local business productivity. Weak water and sanitation infrastructure deters investment. Understanding these links requires both technical competence and socio-economic awareness, confirming that GA 2 demands an integrated knowledge base rather than a narrowly technical one. Engineers who combine quantitative skill with historical and contextual understanding are better equipped to investigate unemployment as a system-level problem and to support development strategies that address root causes rather than only visible symptoms.

4. GA 3: DESIGN

Daniel Walters – 30246970

Engineering design (GA 3) contributes to this investigation by examining how existing systems, structures and technologies affect unemployment and poverty. In South Africa, systems such as public works programmes, renewable energy projects, transport networks, housing developments and agricultural infrastructure are often linked to job creation and economic participation. However, the existence of these systems does not automatically mean that they benefit all communities equally or that their design adequately addresses the structural drivers of unemployment.

Figure 3 illustrates quarter-to-quarter changes in unemployment between Q4 2017 and Q4 2023. The data reveals significant instability during the COVID-19 lockdown period and confirms that unemployment recovery has been slow and incomplete. The fourth quarter of 2023 recorded the fourth consecutive annual increase in fourth-quarter unemployment since the pandemic, demonstrating that existing development systems have not been sufficient to reverse the trajectory (Statistics South Africa, 2024a).

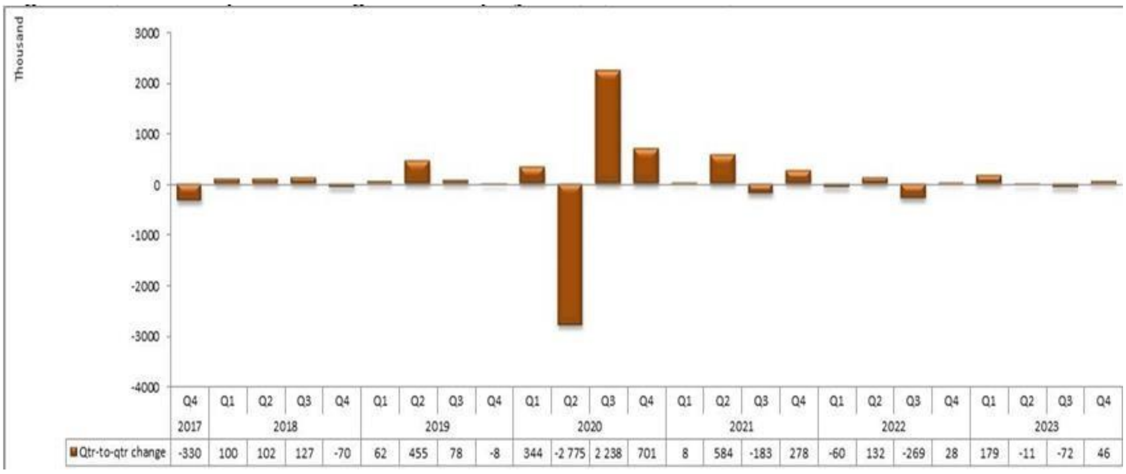


Figure 3: Quarter-to-quarter changes in unemployment in South Africa, Q4 2017 to Q4 2023. (Source: Statistics South Africa, 2024a)

A fundamental design challenge is unequal access. Large-scale infrastructure projects may create employment in some regions while leaving rural areas, informal settlements and townships disconnected from economic centres. The World Bank (2025a) describes South Africa as a dual economy characterised by low growth, high inequality and insufficient job creation, with growth rates too weak to materially reduce poverty or unemployment. From a design perspective, development systems must be evaluated according to accessibility, affordability, maintenance capacity and social benefit – not only technical performance. GA 3 highlights that inclusive engineering design must ensure that benefits reach the communities most affected by exclusion, and that infrastructure investments create sustainable, long-term employment pathways rather than short-term or geographically limited opportunities.

5. GA 4: INVESTIGATION

Diya-Daniel Matulizo – 30647479

Graduate Attribute 4 focuses on systematic investigation, which is essential for building an evidence-based understanding of the relationship between unemployment and poverty. A rigorous investigation requires both quantitative and qualitative data drawn from credible, official sources, and must apply appropriate analytical methods to identify patterns, test relationships and produce supported conclusions.

Key data for this investigation includes unemployment rates, poverty levels, income distribution, access to basic services, education attainment, transport access and regional disparities. Reliable sources include Statistics South Africa, the World Bank, the International Monetary Fund and peer-reviewed research. The Quarterly Labour Force Survey (QLFS) is the primary official source for national and provincial labour-market data in South Africa. Table 1 below summarises the key labour-market indicators from the Q1 2024 QLFS, confirming the scale of the unemployment crisis.

Table 1: Key Labour Market Indicators – South Africa, Q1 2024 (Statistics South Africa, 2024a)

Indicator	Q1 2024 — Official Figure
Official unemployment rate	32.9%
Number of unemployed persons	8.2 million
Expanded unemployment rate	41.9% (includes discouraged work-seekers)
Youth unemployment (ages 15–24)	59.7%
Youth unemployment (ages 25–34)	40.7%
Total employed persons	16.7 million
Graduate unemployment rate	11.8% (vs 39.1% for those without matric)

Data collection methods relevant to this investigation include secondary data analysis of national surveys, census reports, provincial poverty studies and international comparative datasets. Once collected, data should be processed using trend graphs, regional bar charts, comparative tables and GIS-based poverty maps. These tools allow researchers to identify where unemployment and poverty overlap most severely, and to communicate evidence-based findings to technical and non-technical stakeholders alike. This investigation approach moves the analysis beyond general statements, ensuring that conclusions are grounded in verifiable data from the South African context.

6. GA 5: TOOLS, TECHNOLOGIES AND METHODS

Jonathan Standage – 29965772

Tools, technologies and methods play an important role in investigating poverty and unemployment because they enable researchers to identify where disadvantage is concentrated, how people access economic systems, and where infrastructure gaps are most acute. Two categories of tools are particularly relevant to this investigation: spatial analysis technologies for mapping inequality, and digital financial technologies for improving economic participation.

Geographic Information Systems (GIS) and remote sensing are essential tools for analysing spatial inequality. GIS enables researchers to overlay poverty data on settlement patterns, transport networks, proximity to schools and clinics, night-time light intensity and infrastructure coverage. This allows investigators to identify specific areas where multiple dimensions of disadvantage overlap, producing detailed poverty maps that reveal more about the spatial structure of exclusion than national averages can show.

Recent peer-reviewed research demonstrates that remote-sensing transfer learning and satellite-based approaches can support the assessment of economic well-being in contexts where traditional data is delayed or geographically limited (Ni et al., 2025). These tools support evidence-based planning by helping stakeholders identify where infrastructure investment would most improve labour-market access.

Mobile banking and digital financial technologies are also relevant to economic participation. USSD-based platforms and digital wallets allow people to receive social grants, make payments, access savings and conduct financial transactions without requiring a physical bank branch or smartphone.

In South Africa, where many unemployed people live far from formal financial infrastructure, mobile access can reduce transaction costs and support small-scale economic activity. FinMark Trust (2023) confirms that mobile money plays a meaningful role in bridging the gap between formal and informal economic participation. The World Bank (2025b) emphasises that digital financial inclusion can reduce access barriers when systems are affordable, reliable and widely trusted.

However, a critical engineering perspective is essential. GIS depends on current, high-quality data. Mobile banking requires stable connectivity, digital literacy and institutional trust – conditions not equally available across South Africa. GA 5 therefore demonstrates that tools and technologies are most valuable when deployed as part of a broader systems strategy, rather than as isolated responses to the complex, multi-dimensional challenge of poverty and unemployment.

7. GA 6: COMMUNICATION

Gretha du Toit – 30442338

Effective communication is a fundamental engineering skill, particularly when investigating complex social problems such as unemployment and poverty. GA 6 requires that technical and statistical findings be communicated accurately, clearly and in ways accessible to diverse audiences – including policymakers, businesses, communities and the public. In this investigation, communication serves two key purposes: conveying the scale of the problem and enabling evidence-based decision-making across stakeholder groups.

Data visualisation is among the most powerful tools for communicating unemployment data. According to Nediger (2025), data visualisation is the visual presentation of data or information in the form of graphs, charts, infographics or maps, with the goal of identifying patterns and communicating findings clearly and effectively. Across multiple years of quarterly labour-market data and multiple demographic groups, visual tools allow stakeholders to identify structural trends without requiring deep statistical expertise.

Figure 4 shows South Africa's official unemployment rate between 2023 and early 2025, confirming that the rate has remained consistently above 32% with no sustained downward trend. This visual demonstrates that unemployment is structural rather than cyclical: it has not recovered meaningfully from the COVID-19 peak and remains stubbornly resistant to existing interventions (Statistics South Africa, 2024a).

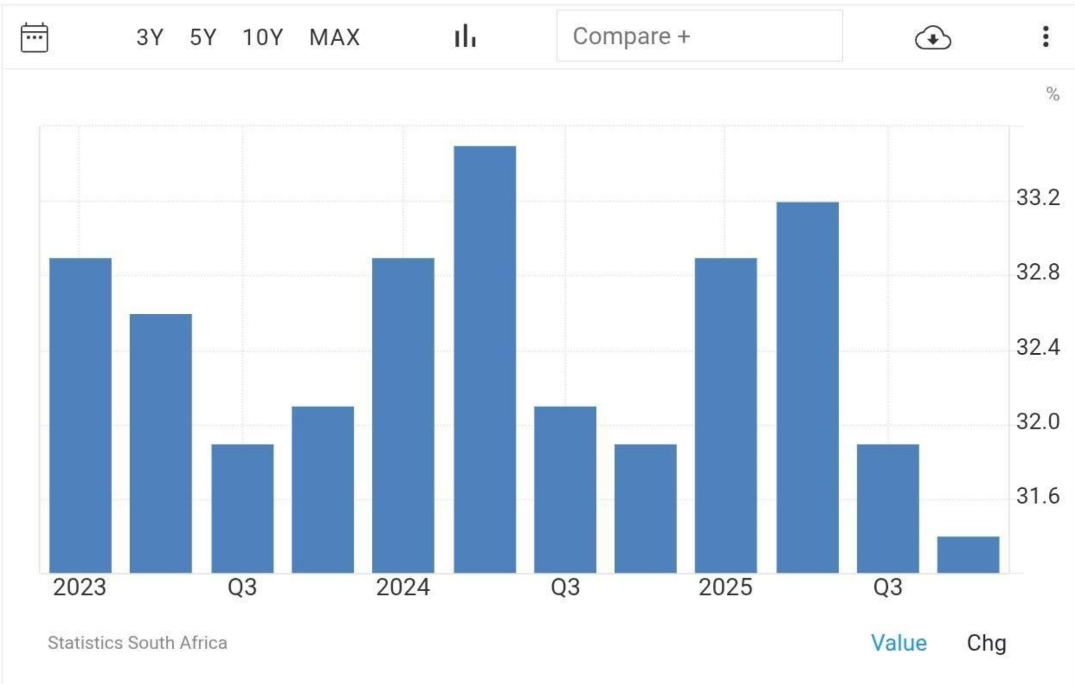
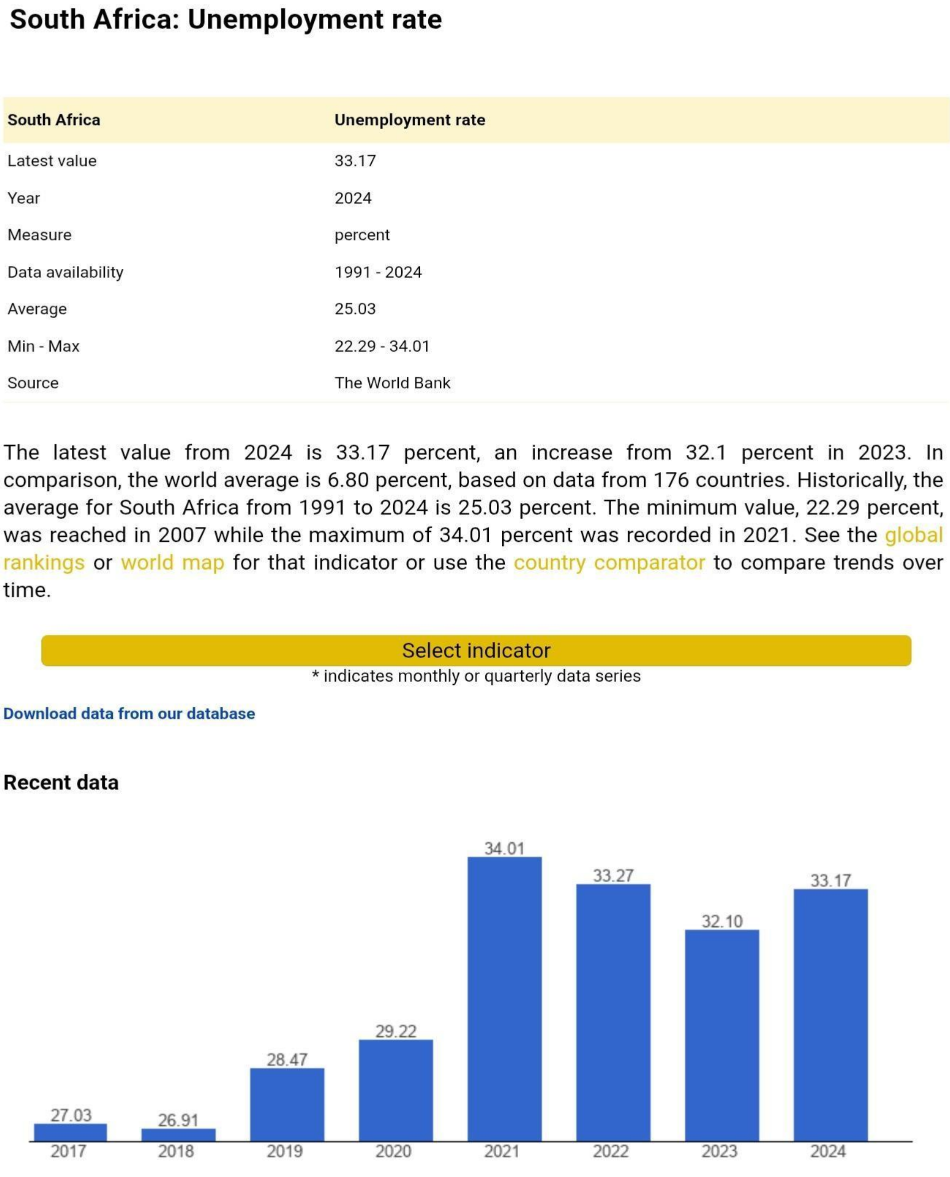


Figure 4: South Africa's unemployment rate, Q1 2023 to Q1 2025. (Source: Statistics South Africa, 2024a)

Figure 5 presents annual unemployment data from 2017 to 2024, showing that the rate increased from approximately 27% in 2017 to over 33% by 2024. The gap between the official rate (32.9%) and the expanded rate (41.9%) in Q1 2024 reveals the significant number of discouraged work-seekers who have left the labour force entirely – a dimension of the crisis often hidden in headline statistics (Statistics South Africa, 2024a; TheGlobalEconomy.com, 2024).



South Africa: Unemployment rate

South Africa	Unemployment rate
Latest value	33.17
Year	2024
Measure	percent
Data availability	1991 - 2024
Average	25.03
Min - Max	22.29 - 34.01
Source	The World Bank

The latest value from 2024 is 33.17 percent, an increase from 32.1 percent in 2023. In comparison, the world average is 6.80 percent, based on data from 176 countries. Historically, the average for South Africa from 1991 to 2024 is 25.03 percent. The minimum value, 22.29 percent, was reached in 2007 while the maximum of 34.01 percent was recorded in 2021. See the global rankings or world map for that indicator or use the country comparator to compare trends over time.

Select indicator
* indicates monthly or quarterly data series

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Figure 5: South Africa's unemployment rate, 2017–2024, with summary statistics. (Source: TheGlobalEconomy.com, 2024, using World Bank data)

Strong communication between stakeholders – through reliable data, clear visualisations and contextually sensitive messaging – is essential for building the shared understanding needed to address unemployment effectively. GA 6 confirms that the ability to translate complex findings into understandable arguments is not peripheral to engineering, but central to its social impact.

8. GA 7: ENGINEERING IN THE WORLD

Abner Paul – 30651883

Graduate Attribute 7 requires engineers to understand the broader social, economic, health, legal and environmental dimensions of engineering activity. Applied to SDG 1 and unemployment in South Africa, GA 7 demands a holistic stakeholder analysis that moves beyond individual employment statistics to examine how unemployment affects multiple interconnected systems simultaneously.

The primary stakeholders in this investigation include unemployed individuals, low-income households, rural and township communities, businesses, government institutions, schools, healthcare providers, transport operators and engineering professionals.

Each stakeholder experiences the consequences of unemployment differently, but all are affected by the broader consequences of persistent labour-market exclusion. Unemployed persons face direct income loss, while businesses experience reduced consumer demand and a weakening skills pipeline. Government institutions face growing pressure on social grants and public services. Schools and healthcare providers operate in communities where financial stress and poor nutrition undermine learning and health outcomes.

Socially, unemployment increases dependency, deepens inequality and reduces quality of life across generations. Youth unemployment is especially significant: with 59.7% of those aged 15–24 unemployed in Q1 2024, an entire generation faces reduced access to skills development and economic independence (Statistics South Africa, 2024a). Long-term social consequences include eroded intergenerational mobility, increased community instability and the entrenchment of spatial poverty in disadvantaged areas.

Economically, high unemployment reduces household income and weakens consumer spending. The World Bank (2025a) notes that South Africa's low growth and insufficient job creation remain central development challenges. When large numbers of people cannot earn income, aggregate demand falls, local businesses experience lower revenues and investment in affected areas contracts.

From a health perspective, long-term unemployment is associated with poor nutrition, elevated psychological stress, reduced access to healthcare and increased risk of non-communicable diseases (World Health Organization, 2023).

Legally and institutionally, labour laws, procurement systems, education policy and infrastructure investment decisions all determine whether job opportunities are created or restricted. GA 7 demonstrates that unemployment is a multidimensional problem that requires holistic investigation and integrated responses.

9. GA 10: ETHICS AND PROFESSIONALISM

Corne Meintjies – 305381140

Graduate Attribute 10 focuses on professional responsibility, ethical conduct and sound judgement. In the context of SDG 1 and unemployment, GA 10 requires engineers to recognise how their decisions influence whether development projects create genuine social benefit or reinforce existing inequalities. Ethical failure in engineering practice has direct consequences for unemployment and poverty: corruption in procurement, substandard construction, inadequate maintenance and working beyond one's competence all reduce the employment-generating and poverty-reducing potential of infrastructure investment.

According to the Engineering Council of South Africa, engineers must demonstrate critical awareness of the need to act professionally and ethically, and must take responsibility for the consequences of their decisions on society (ECSA, 2023). This obligation is especially important in South Africa, where the direct link between infrastructure quality, access to economic opportunity and poverty is well established. When infrastructure is delayed, poorly built or inaccessible to the communities it was intended to serve, the benefits of development investment are lost or significantly diminished. In a country where over 8 million people are unemployed and the majority of the population lives under financial stress, poorly governed engineering projects represent not only a waste of resources but a failure of professional responsibility.

Professional engineering practice in this context requires accountability, integrity, competence and genuine commitment to the public good. Engineers must ensure that projects are technically sound, socially accessible, economically sustainable and aligned with community needs. The mechanism between unethical practice and unemployment is clear: corruption delays infrastructure; delayed infrastructure reduces business confidence; reduced confidence limits hiring; and poor-quality infrastructure increases maintenance costs while failing to connect communities to economic opportunity.

ECSA's codes of conduct require engineers to act with integrity, prioritise public safety and promote the public good (ECSA, 2023). These principles align directly with SDG 1, which targets the elimination of poverty and universal access to economic resources (United Nations, 2024). GA 10 therefore demonstrates that ethical engineering practice is not a separate concern from poverty reduction – it is an integral precondition for it.

10. CONCLUSION

Group

This report investigated poverty in South Africa by examining unemployment as its primary systemic driver. Through the Engineering Graduate Attributes, the investigation demonstrated that unemployment is not a simple or isolated labour-market problem. It is a complex, historically rooted and structurally sustained challenge shaped by unequal infrastructure, spatial exclusion, limited education access, transport barriers, weak economic growth and gaps in ethical governance.

The key findings are clear. Unemployment in South Africa is persistent and severe: official unemployment of 32.9%, with 8.2 million people unemployed and youth unemployment at 59.7% in Q1 2024, represents a structural condition rather than a temporary fluctuation (Statistics South Africa, 2024a). Unemployment affects more than income: it weakens food security, health, skills development, community stability and the capacity of local markets. Existing infrastructure systems have not been designed or distributed in ways that consistently reach the most affected communities. Digital and spatial tools can strengthen investigation and response, but require enabling conditions that are not equally available across South Africa. And ethical engineering practice is not peripheral to development – it is foundational to ensuring that investment reaches communities most in need.

From an engineering perspective, the investigation highlights that inclusive design, rigorous data analysis, clear communication, appropriate technology use and professional integrity are all essential. Engineers cannot resolve unemployment alone, but they can contribute to systems that reduce spatial exclusion, improve infrastructure access, support skills development and strengthen local economic participation. Addressing unemployment is essential for progress toward SDG 1: stronger access to work supports stronger communities, stronger markets and more sustainable long-term development. This investigation demonstrates that engineering, applied responsibly and evidence-based, has a meaningful and indispensable role in that process.

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